

Sangbeom Park

AI RESEARCH ENGINEER

Seoul, Republic of Korea

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Work Experience

ATLO

CTO, FULL-TIME

- Developed and deployed an AI social robot service using LLMs and multimodal pipelines

Seoul, S.Korea

Jun. 2024 – Apr. 2025

NAVER Cloud

MACHINE LEARNING ENGINEER, INTERN

- Developed NewsBot, a personalized news delivery system powered by HyperCLOVA X

Gyeonggi, S.Korea

Mar. 2024 – Jun. 2024

NAVER LABS

MACHINE LEARNING ENGINEER, INTERN

- Implemented unsupervised skill learning to develop safe and diverse quadruped locomotion behaviors (Mentor: Taeyoon Lee)

Gyeonggi, S.Korea

Mar. 2023 – Sep. 2023

Education

Korea University

MASTER OF SCIENCE IN ARTIFICIAL INTELLIGENCE

- Thesis: Learning Diverse Manipulation Skills for Effective Semi-Autonomous Teleoperation
- Cumulative GPA: 4.42/4.5 | Credit: 40
- Supervisor: Sungjoon Choi

Seoul, S.Korea

Sep. 2021 – Feb. 2025

Dongguk University

BACHELOR OF SCIENCE IN ELECTRICAL AND ELECTRONICS ENGINEERING & MINOR IN INTELLIGENT ROBOTICS ENGINEERING

- EE GPA: 4.09/4.5 | IR GPA: 4.43/4.5 | Credit: 149
- Leave of absence for military service in KATUSA: Nov. 2016 – Aug. 2018

Seoul, S.Korea

Mar. 2015 – Aug. 2021

Publications

Quality-Diversity based Semi-Autonomous Teleoperation using Reinforcement Learning

SANGBEOM PARK, TAERIM YOON, JOONHYUNG LEE, SUNGHYUN PARK, AND SUNGJOON CHOI

- [Project Page](#) | [Paper](#) | [Code](#) | [Poster](#)
- Presented on Physical Human-Robot Interaction at *ICRA 2024 Workshop*
- Listed in COMPUTER SCIENCE, ARTIFICIAL INTELLIGENCE & NEUROSCIENCES (*Impact Factor: 7.8, JCR Top 10%*)

Neural Networks

Nov. 2024

Visual Preference Inference: An Image Sequence-Based Preference Reasoning in Tabletop Object Manipulation

JOONHYUNG LEE, SANGBEOM PARK, YONGIN KWON, JEMIN LEE, SUNGJOON CHOI

- [Project Page](#) | [Paper](#) | [Code](#) | [Video](#) | [Poster](#)
- Presented on Vision-Language Models for Navigation and Manipulation at *ICRA 2024 Workshop*

IROS

Oct. 2024

SPOTS: Stable Placement of Objects with Reasoning in Semi-Autonomous Teleoperation Systems

JOONHYUNG LEE, SANGBEOM PARK, JEONGEUN PARK, KYUNGJAE LEE, AND SUNGJOON CHOI

- [Project Page](#) | [Paper](#) | [Code](#) | [Video](#)

ICRA

May. 2024

CLARA: Classifying and Disambiguating User Commands for Reliable Interactive Robotic Agents

JEONGEUN PARK, SEUNGWON LIM, JOONHYUNG LEE, SANGBEOM PARK, MINSUK CHANG, YOUNGJAE YU, AND SUNGJOON CHOI

- [Project Page](#) | [Paper](#) | [Code](#) | [Video](#)

RA-L

Dec. 2023

Semi-Autonomous Teleoperation via Learning Non-Prehensile Manipulation Skills

SANGBEOM PARK, YOONBYUNG CHAI, SUNGHYUN PARK, JEONGEUN PARK, KYUNGJAE LEE AND SUNGJOON CHOI

- [Paper](#) | [Code](#) | [Video](#)

ICRA

May. 2022

Projects

Interactive Multimodal Agent for Emotion-Aware Dialogue

ATLO

AI RESEARCH ENGINEER & CTO

Jun. 2024 - Apr. 2025

- Launched and operated a web-based dialogue agent service (*Atto*) and iOS app *Ditto*.
- Designed an asynchronous multimodal interaction pipeline.
- Implemented a RAG system using LangChain and LLMs with prompt engineering.
- Built full-stack web and iOS apps enabling multimodal chat with the agent.
- Secured 110M KRW seed funding at a 1.1B KRW valuation from Strong Ventures.
- Achieved 54,000+ conversation sessions in four months.

AI News Curation System

Naver Cloud

MACHINE LEARNING ENGINEER & INTERN

Mar. 2024 - Jun. 2024

- Built backend infrastructure with AWS, Docker, and MySQL.
- Generated labeled data using LLMs and fine-tuned HyperCLOVA X for classification.
- Applied embedding similarity and reranking to filter and prioritize articles.
- Deployed as a chatbot, delivering curated news to 900+ employees.

Safe and Low-Noise Locomotion Skill Learning

NAVER LABS

MACHINE LEARNING ENGINEER & INTERN

Mar. 2023 - Sep. 2023

- Designed a Safe Critic Network to suppress unsafe/noisy gaits.
- Used Hierarchical RL to learn diverse walking behaviors with safety-aware policies.
- Improved sim-to-real transfer via domain randomization in Isaac Gym.
- Developed a real-time Python-C++ control pipeline using the LCM communication framework.
- Reduced walking noise by 24%, validated in simulation and real hardware.

Automation Maintenance using Mobile Manipulator

SAMSUNG

M.S. RESEARCH PROJECT

Mar. 2021 - Sep. 2023

- Created simulation environments in Unity and Gazebo for autonomous maintenance tasks.
- Developed navigation and manipulation capabilities for mobile manipulators in factory settings.

Presentations

Physical Human-Robot Interaction

Yokohama, Japan

CONFERENCE PRESENTATIONS

May, 2024

- Poster presentation at *IEEE International Conference on Robotics and Automation (ICRA)*

Learning X in Physics-based Simulators

Seoul, S.Korea

TECH TALK

Mar. 2024

- Invited talk at *CINAMON, AI Research Group*

Shared-Autonomy for Robotics

Gyeonggi, S.Korea

TECH TALK

Feb. 2023

- Invited talk at *NAVER LABS, Robot Dynamics & Control Group*

Telerobotics

Philadelphia, U.S.A

CONFERENCE PRESENTATIONS

May, 2022

- Oral presentation at *IEEE International Conference on Robotics and Automation (ICRA)*

Patent

2023 **10-2023-0125345**, Semi-Autonomous Teleoperation via Learning Manipulation

Korea

Skills

Programming	Python, C++, C#, Java
Physics Simulation	Isaac Sim, MuJoCo, Unity, Gazebo
Realworld Robots	<u>UR5e</u> , <u>Mini-Cheetah</u> , <u>Franka Panda</u> , <u>Pioneer 3-DX</u> , <u>Xycar</u>
DevOps	Amazon ECS, Docker
Languages	Korean (Native), English (Fluent)